

Entity Framework Core

1) Model Class : class Emp.

↳ Attributes : Table, Column, Key

2) DAL :-

DbContext Class : in this, we need to create a DbContext object which is mapped with Database table.



migration :- Simple C# Classes,
Decides what should be the
database structure as per
the model class.

When you run this migration —
It generates database tables
accordingly.

⇒ add - migration "Name Of C# Class"
⇒ update - database

Microsoft - Entity Framework Core

⇒ Provides DbContext class

for Database communication.

- ✓ read the connection string
- ✓ Create query, fire query,
- ✓ Open and close the connection
- ✓ send / fetch the data from Db - .NET 7, .NET to db.

Microsoft - Entity Framework Core. SqlServer

⇒ Database Provider

Microsoft. Entity Framework Core. Tools

⇒ They're primarily used to manage migrations and to scaffold a Db Context.

Microsoft. Entity Framework Core. Design

⇒ contains all the design time logic for Entity Flw Core.

onConfiguring → override

↳ We were binding a dependency on a Service which can read app settings. json file and can get connection string for DbContext at runtime.

DBSet < Emp > emps prop

- dbContent.emps.ToList()

↓

Select * from Emp.

GOP

Events & Delegate

Collection, Generic, Generic Collection

File 20, Serialization

Exception Handling

Reflection

ADO.NET:- Connected
Disconnected
EF Core

Threading → 4 = MVC
Realtime → MVC

- TPZ

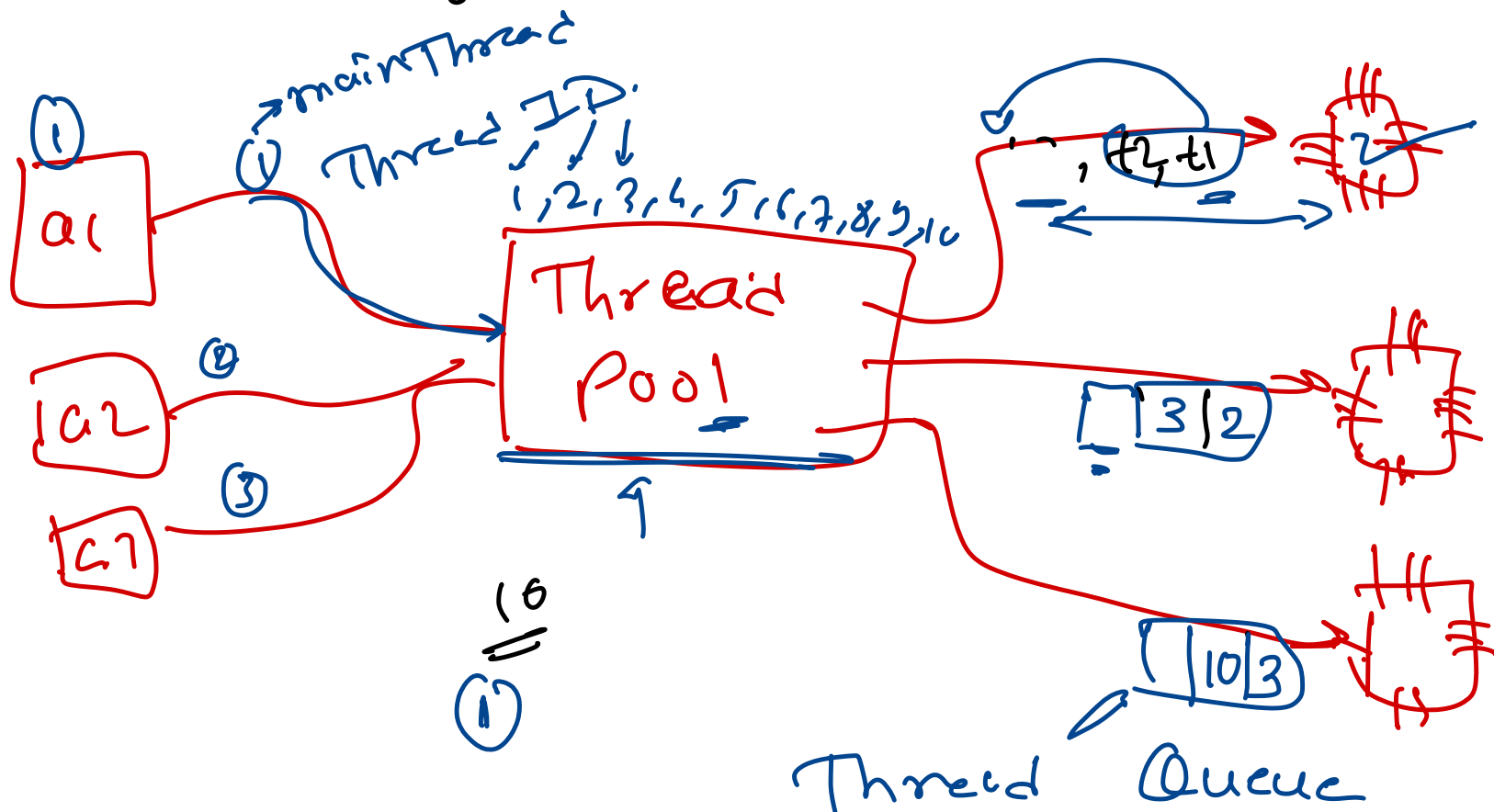
- Parallel - Prog.

Threading :-

System.Threading namespace

class Thread.

↓
delegate Thread Start



Task

lighter than Thread

App.

Global Task Queue

1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Thread Queue

3, 2, 1

$n = 3 \times \underline{\underline{tk3, tk2, tk1}}$

Thread Pool

1, 2, 3, 4, (1-10)

6, 5, 4

9, 8, 7

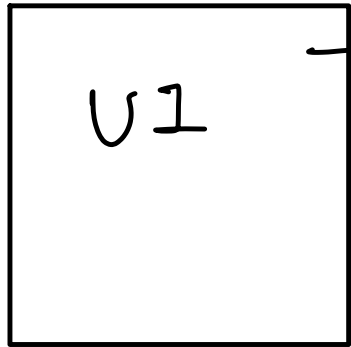
CLP can add
programmatically
create Threads
to Thread Pool

(1-10)
Thread

Client - Server Architecture

Browser

WebServer



http://www.Mitra.com

B. Logic
PHP, Node,
Java, .NET

IP Addr.

html, CSS, JS, JSON, XML

DNS Server

IP Addr

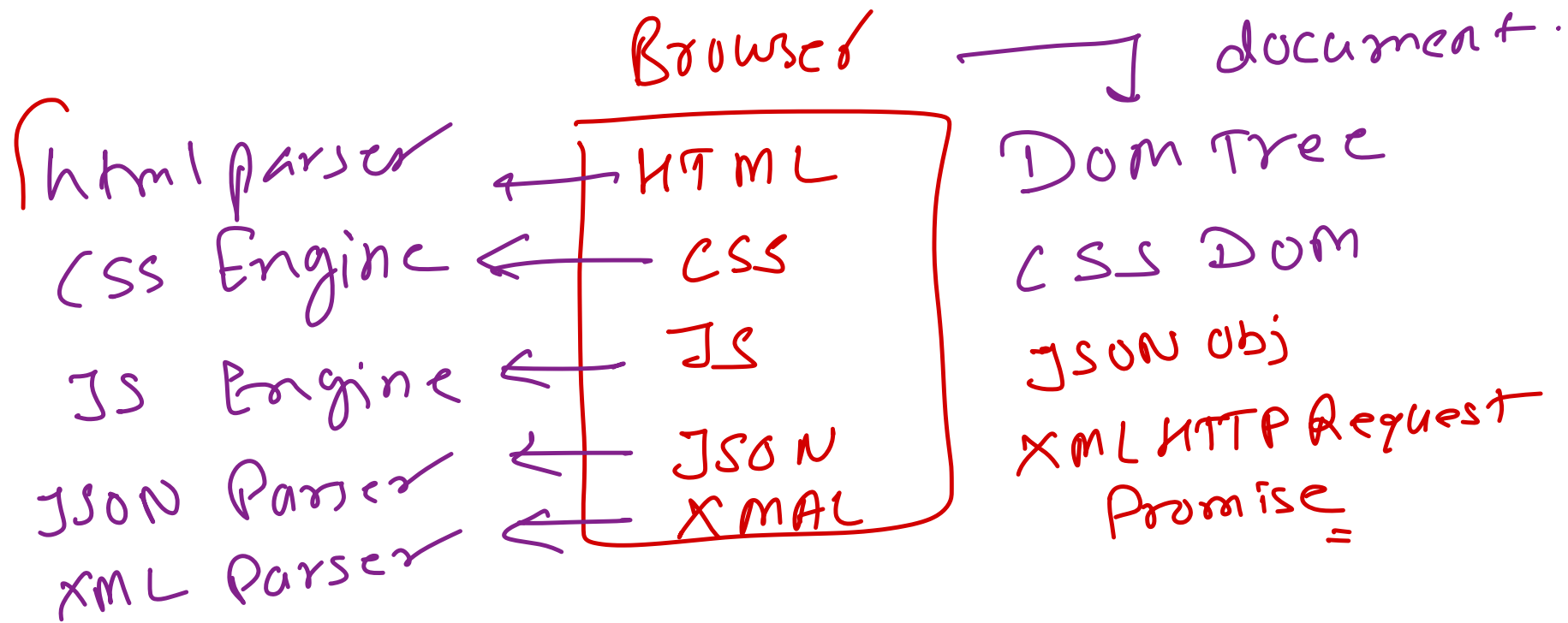
IP Addr

Mitra
Amazon

Dir :- OnlineShop

678.256.23.28

IP Addr.



<p> ABC </p> ← Browser

html + css + js

Server App^m → Render → html, css,
js, JSON,
emit
generate
plaintext -

UN

PWD

name

"UN = Admin, PWD = 1234"

"UN = Admin"

"PWD = 1234"

["UN", "Admin"]

["PWD" = "1234"]

["key": "UN", "Value": "Admin"]

["key": "PWD", "Value": "1234"]

validation(
string UN,
string PWD)

{
}

Client side
div

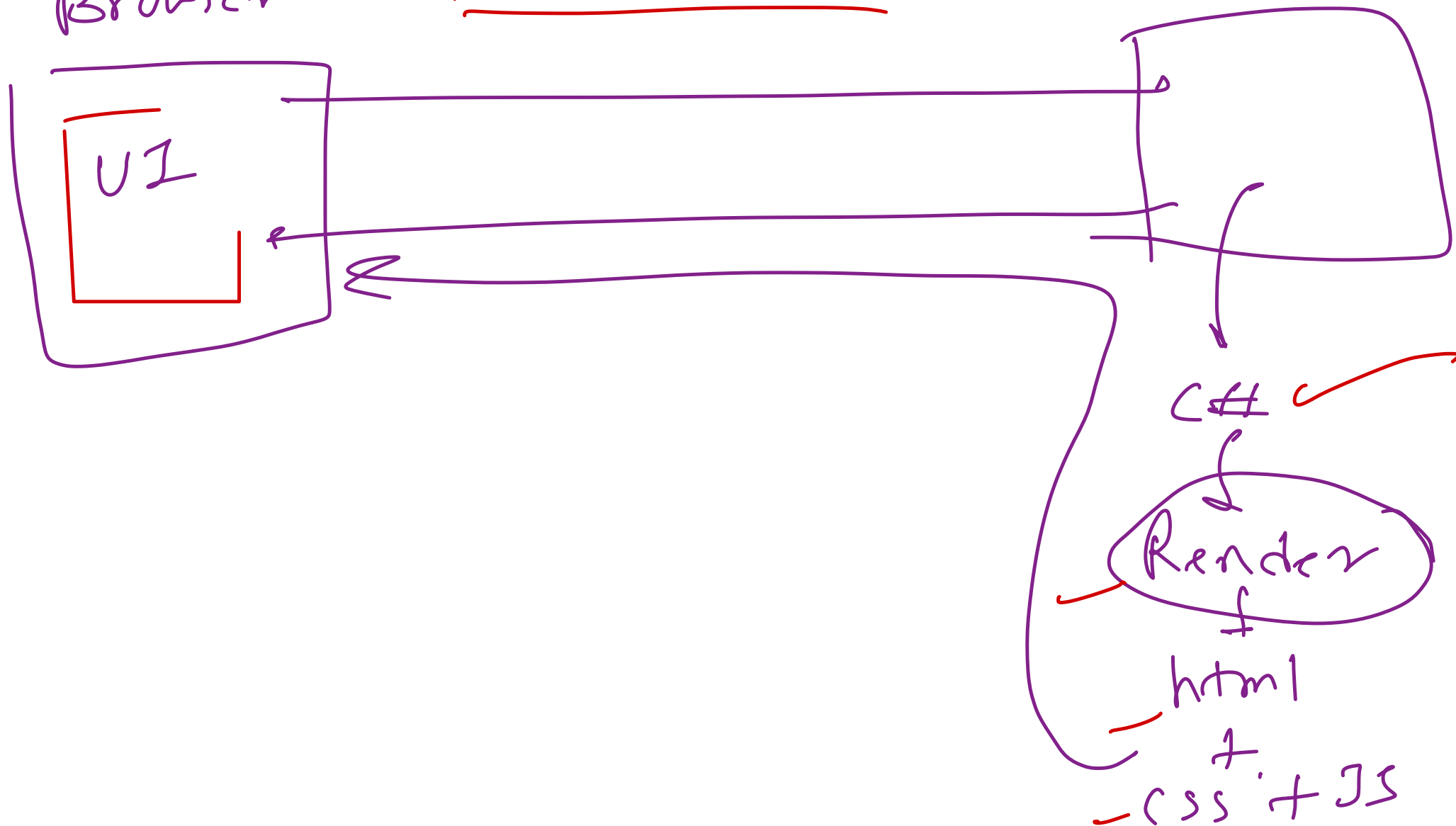
Server side
div

Input type = text
name = "UN"

Class Asp. Web Appⁿ

Browser

Server



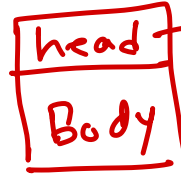
Asp.Net Webform App.ⁿ

Client
Browser

http://127.0.0.1:5700/index.html/.asp

ITS

HTTP Request Packet



URL
method = GET

80

HTTP Pipeline

http
life cycle

Receive & Unload
the http Request Packet

①

HTTP Content obj.
obj. URL = "127.0.0.1/index.html"
obj. HTTPMethod = "GET"
obj. response = class HTTPResponse
obj. request = class HTTPRequest

②

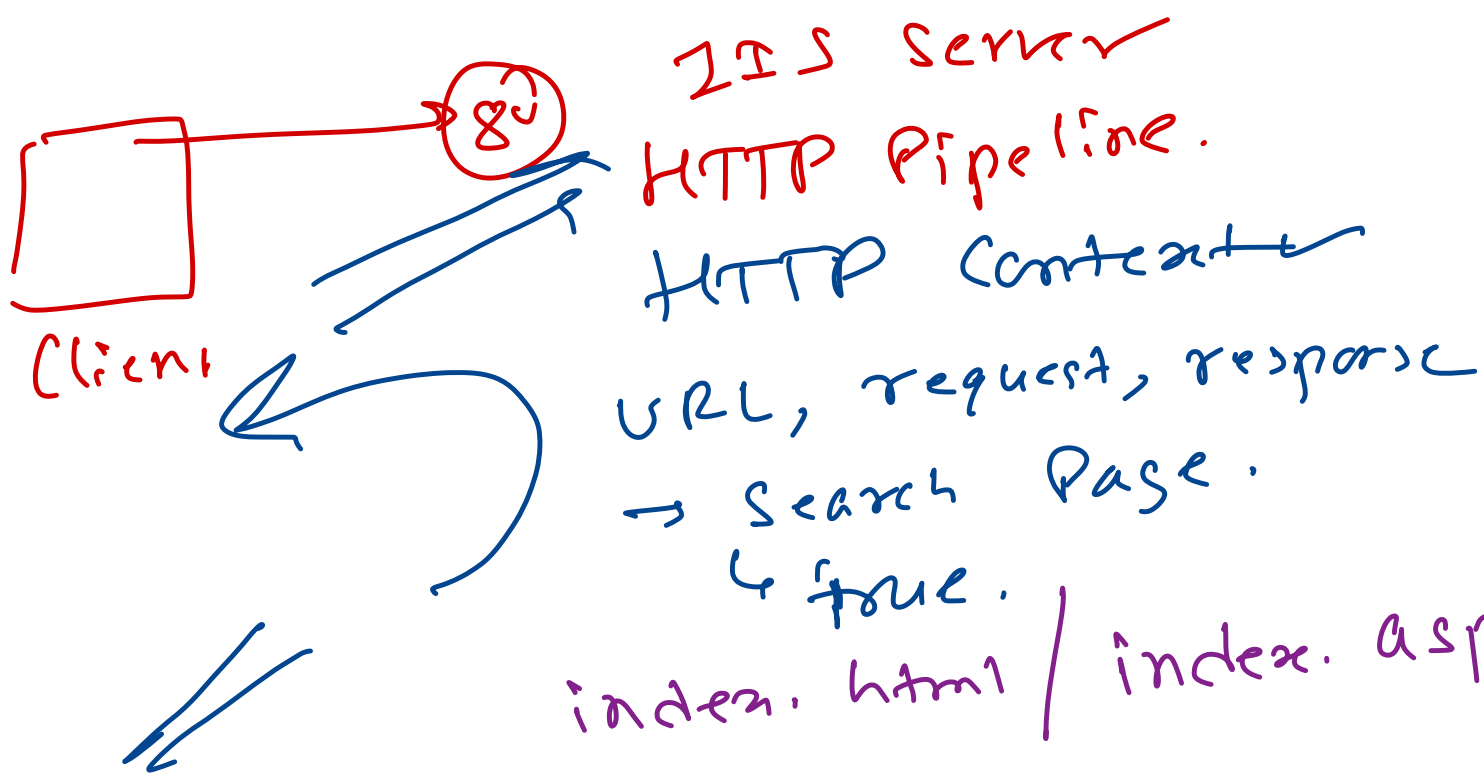
URL = myAppDir
mitra.com = path of
true
search of page
path/index.html
true

my App App.ⁿ life cycle

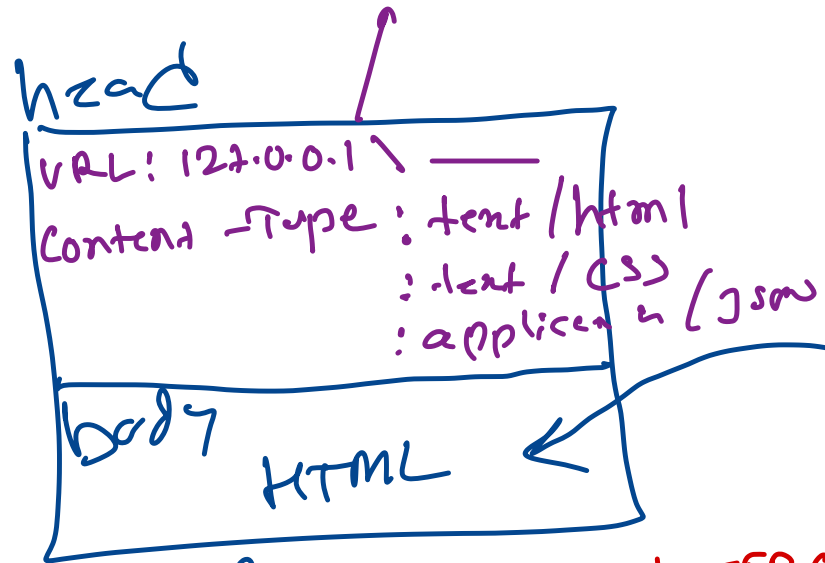
Page Controller Pattern.

Front Page Pattern
↳ page physical path
load search.

↳ 404 Not R



[HTTP is stateless]
close & clean
http pipeline
Send)



HTTP Response
Pack



Surrogate Process

mscorlib.dll

.aspx

aspnet_isapi.dll

CLR

myApp.exe

.asp

aspnet.dll

myApp Pipeline

- Init() ✓
- PageLoad() ✓
- btn_click()

Render()

↳ ^{blend} execute C# and generate html

HTTPContext.response.Add(html)

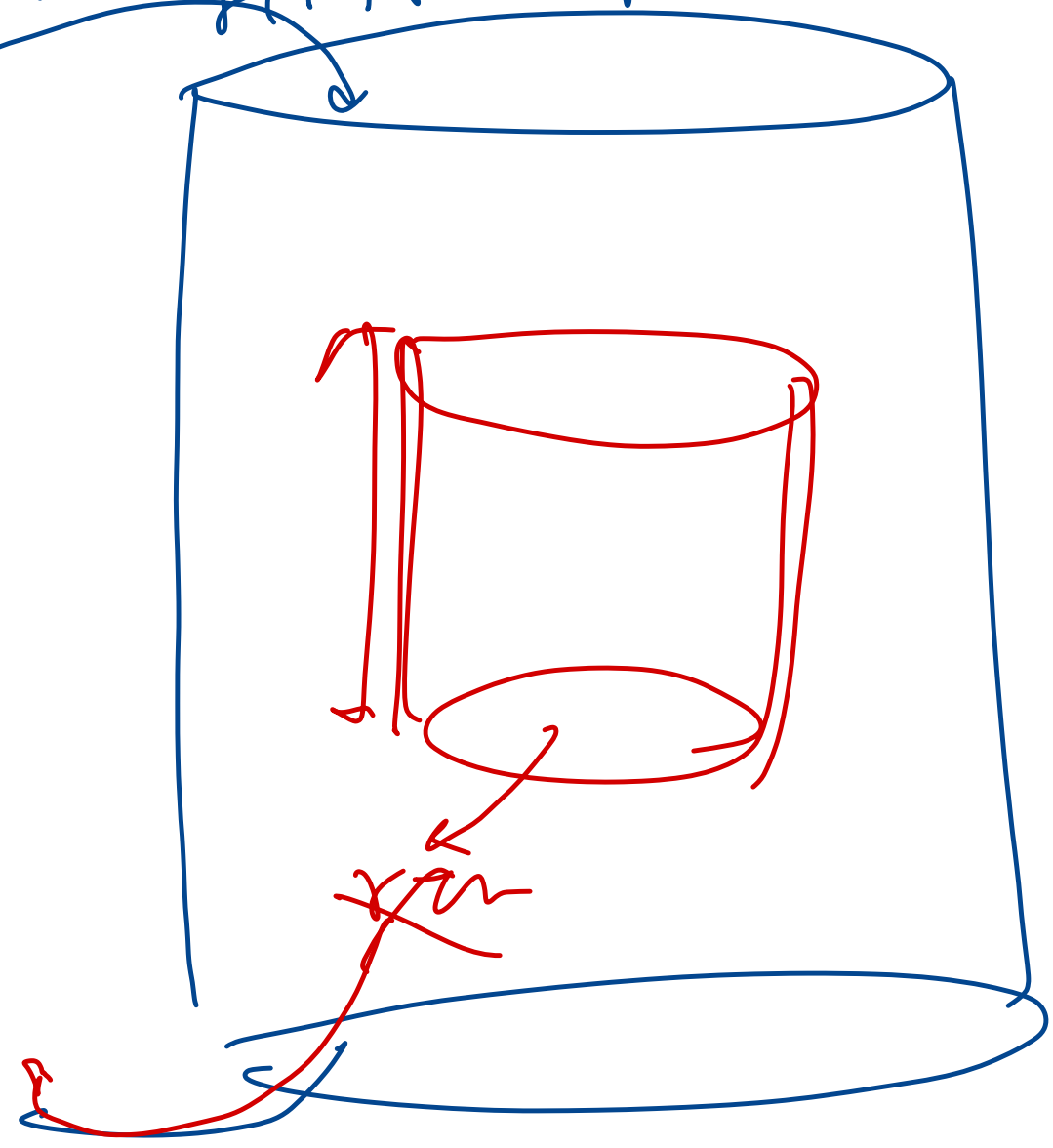
All C# code memory — clean GC

→ Clean CLR, clean isapi.dll

Create / Generate HTTP Response Packet

Request Pack

HTTP pipeline



HTTP Response packet

Asp. Net WebForm Web Appⁿ

↳ No controll over html Generation

↳ Since HTTP is a stateless protocol

↳ Pages were bulky bcoz of duplicate data.

UN:

PWD:

Duplicate copy of Page. ∴ Page, ViewState

P/W Asp.Net MVC Web App'n

a new architectural design

framework based on

Model - View - Controller /

MVC design pattern based.

URL : <http://www.bvs.com/Home/Index>

Search = HomeController

Front Controller Pattern.

Asp. Net
IIS

MVC

→ 1.0
→ 2.0
→ 2.0
→ 4.0
→ 5.0

2015

. Net oper-sum

Asp. Net MVC Core

→ 5.0 : 1.0

2.0

3.0

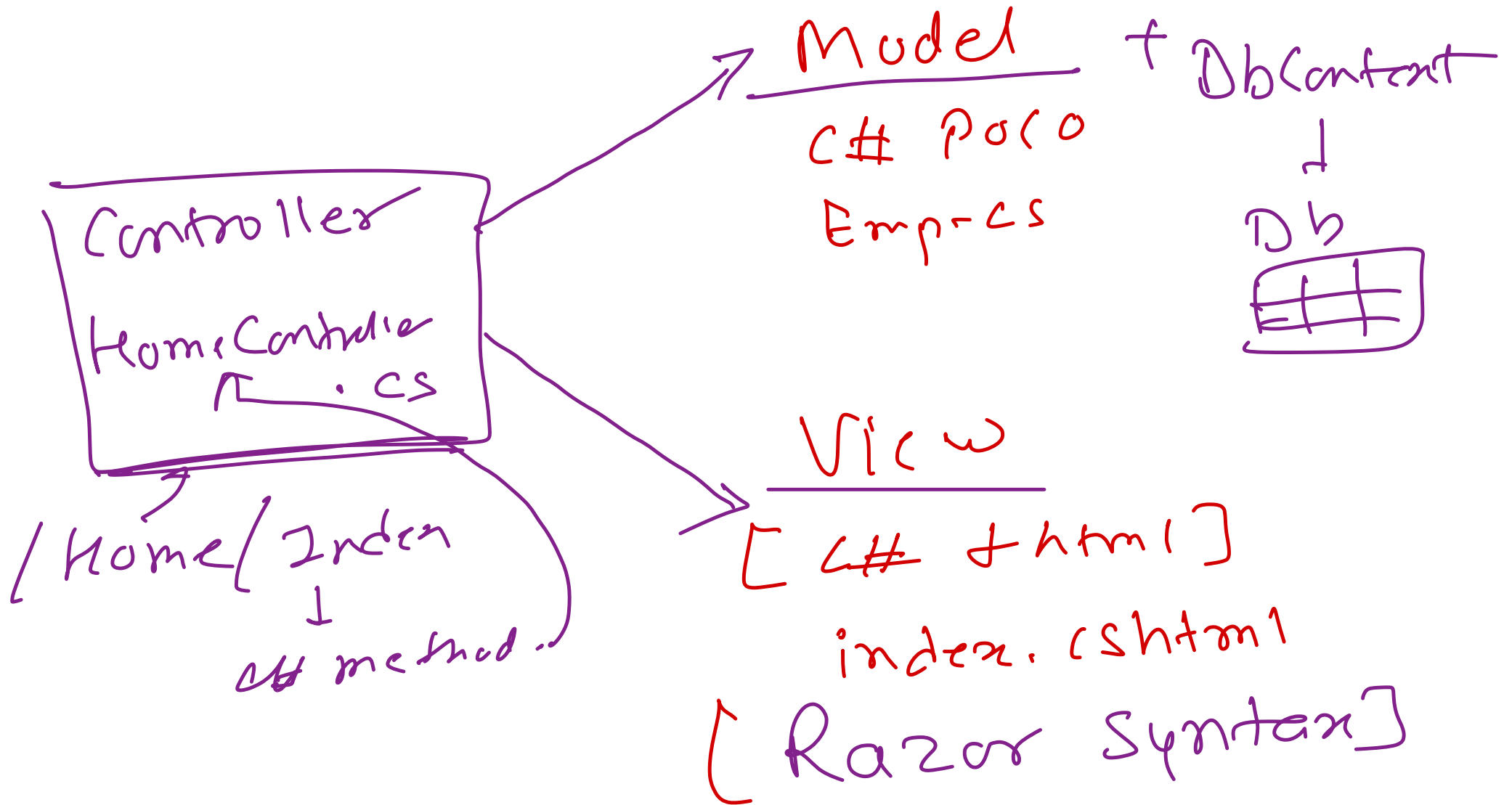
!

6.0

Open-Source

IIS

kernel

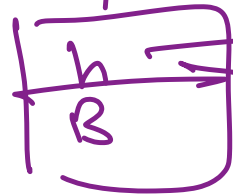


Client
Browser

http://www.brs.com/Home/Index

IIS / kestrel

HTTP Request Packet



URL
Method: Get.

HTTP Pipeline

URL, resp, resp
↳ HTTP Content

URL = "brs.com"

↓
map.

application directory
path

URL = "\Home \Index"

Front Controller Pattern

"\" = "Home", "Index"

["controller" : "Home" + "controller"
: "Home Controller"

"action" : "Index"]

URL: pattern

- ② \ Home \ Index ✓
- ② \ Home \ About ✓
- ② \ Admin \ Dashboard ✓
- ⇒ ④ \ Demo \ login \ unm \ pwd
- ③ \ EmpData \ Update \ id

Routing Middleware

Routing Table

\ controller \ action	✓
\ controller \ action \ <u>unm</u> \ pwd	✓
\ contr \ action \ id	✓
\ contr \ action \ id ?	✓